

Compliance & Certification 101

CE • FCC • BIS • UL • RoHS • REACH

What each means, when you need it, what it costs

Audience: Engineering Students & Startup Founders

Standards • Bodies • Cost • Schedule • Practical paths

Why Compliance Matters

Can't sell without it

Legal requirement

No certs = customs seizes the shipment.

Buyer demand

Retail chains require formal compliance documents.

Safety

Real risk reduction — fires, shocks, RF interference.

Brand

Compliance mark adds credibility.

Liability

Limits your exposure if something goes wrong.

Market access

Different cert per market / region.

Standards by Region

What you need where

Region	Mark	Scope	Approx cost (₹)
India	BIS / WPC	Safety + RF + (some) EMC	1-5 L
EU	CE	Safety + EMC + RoHS + REACH	3-15 L
USA	FCC	RF + EMC	3-10 L
USA / Can	UL / cUL	Safety (electrical/fire)	5-20 L
UK	UKCA	Equivalent to CE post-Brexit	3-10 L
China	CCC, SRRC	Safety + RF	3-12 L
Japan	PSE, MIC	Safety + RF	4-10 L
Australia	RCM	Safety + RF + EMC	3-8 L
Korea	KC	Safety + RF + EMC	3-8 L

What Compliance Covers

Topics tested

Electrical safety

Insulation, creepage, fault current.

EMC emissions

Conducted & radiated at various bands.

EMC immunity

ESD, surge, EFT, radiated susceptibility.

RF emissions

Wireless / Wi-Fi / BT on specific bands.

Substances (RoHS)

Lead, mercury, Cd, Cr-VI, phthalates, PBBs.

Energy efficiency

Standby, off-mode power consumption.

CE Marking (Europe)

Self-declaration with technical file

- ▶ CE = Conformité Européenne; mandatory for sale in EU
- ▶ Self-declaration of compliance (DoC) — you sign for it
- ▶ Backed by a Technical Construction File: schematic, BOM, test reports
- ▶ Must comply with applicable EU directives: LVD, EMC, RED, RoHS, REACH
- ▶ Most products: Low-Voltage Directive (LVD) + EMC Directive
- ▶ RED (Radio Equipment Directive) — anything wireless
- ▶ Tested by Notified Body OR self-tested if standards permit
- ▶ 10-year file retention required

FCC (USA)

Verification, Declaration, or Certification

- ▶ FCC Part 15 — unintentional radiators (most electronics)
- ▶ Three paths:
 - Verification — supplier tests; rare today
 - SDoC (Supplier Declaration of Conformity) — most consumer electronics
 - Certification — required for intentional radiators (Wi-Fi, BLE, etc.)
- ▶ Class A (commercial) vs Class B (consumer) — Class B is stricter
- ▶ Each wireless module needs an FCC ID; modular approval saves time
- ▶ Test at an accredited lab; file with TCB (Telecom Certification Body)

BIS / WPC (India)

Indian certification landscape

- ▶ BIS — Bureau of Indian Standards (safety, RoHS, batteries)
- ▶ WPC — Wireless Planning Commission (wireless approvals)
- ▶ TEC — Telecom Engineering Centre (telecom products)
- ▶ CRS — Compulsory Registration Scheme (electronics — IS 13252, etc.)
- ▶ ETA — Equipment Type Approval (wireless modules)
- ▶ Pre-certified modules (ESP32, nRF52) drastically reduce certification effort
- ▶ Lab tests take 4-12 weeks; allow 2-3 weeks buffer for re-tests
- ▶ Indian labs: Centre for Battery Engineering, SAMEER, ICAT, ERTL

UL / TUV / Intertek — Safety Certs

Listed (full) vs Recognized (component)

- ▶ UL Listed = full product approval (you can sell to consumers)
- ▶ UL Recognized = component within a UL-listed end product
- ▶ Common standards: UL 60950, UL 62368 (info-tech), UL 1310 (PSU)
- ▶ Tests cover: temperature rise, fault current, insulation, fire resistance
- ▶ TUV (German) and Intertek (UK) are equivalent global bodies
- ▶ UL marking on the product is mandatory for sale in many B2B contexts
- ▶ Process: design review → in-lab tests → factory audit → ongoing inspections
- ▶ Cost: ₹5-20 L for a typical product

RoHS, REACH, WEEE — Substance Restrictions

Environmental and health regulations

RoHS

Restricts 10 substances:
lead, mercury, cadmium,
Cr-VI, PBB, PBDE, etc.

REACH

EU chemical registry —
SVHC (Substances of Very
High Concern) tracking.

WEEE

Producer responsibility
for end-of-life electronics.

ELV

End-of-Life Vehicles —
automotive-specific.

Conflict

Tin, tantalum, tungsten,
gold — supply-chain audit.

Halogen-free

Bromine + chlorine
in PCB resin → flame retardant.

Pre-Certified Modules — The Shortcut

Buy modules with cert done

- ▶ Wireless modules (ESP32, nRF52, Quectel, SIMCom) ship with FCC/CE/BIS approvals
- ▶ Saves ~70-90% of compliance cost & schedule
- ▶ Module's FCC ID can be reused in your product
- ▶ Must follow vendor's integration rules (antenna keep-out, etc.)
- ▶ Module pre-cert covers RF; you still need EMC for the whole product
- ▶ Trade-off: higher unit cost, but easier compliance
- ▶ Common modules: ESP32-WROOM, BLE modules from Murata, cellular modules from Quectel/u-blox

Cost & Schedule Reality

Plan early — these eat 3-6 months

Activity	Cost (₹)	Time	Notes
Pre-compliance scan	20-80k	1-2 weeks	Find issues cheap
EMC test (full)	2-5 L	3-6 weeks	Plus retest budget
Safety test	3-8 L	4-8 weeks	Pre-test fixtures take time
RF cert (per region)	2-10 L	4-12 weeks	India + US + EU = 3 cycles
RoHS test	10-30k	2 weeks	Per material change
Total typical product	10-30 L	3-6 months	Include retest buffer

Tip

Budget 1.5× initial estimate. Re-tests are common — one failure usually means ₹50k-2L extra.

What the Lab Needs From You

Documentation package

- ▶ User manual & marketing material
- ▶ Block diagram & schematic
- ▶ Photos of the product (top, bottom, internal)
- ▶ BOM with manufacturers
- ▶ Test fixture if specialised
- ▶ Firmware mode for compliance (e.g. transmit forever for RF)
- ▶ Working samples — typically 3-5 units
- ▶ Antenna gain / directivity data (for RF cert)

Common Compliance Mistakes

What real start-ups learn the hard way

- ▶ Not budgeting for compliance — surprise ₹10 L bill at launch
- ▶ Designing without certification timeline — slips 6 months at the end
- ▶ Skipping pre-compliance — failing full cert costs 3-5×
- ▶ Wireless without pre-cert module — full RF cert for every region
- ▶ Using grey-market parts — supply chain audit fails
- ▶ Ignoring RoHS / REACH — buyers refuse to source
- ▶ Compliance marks missing on product — customs delay
- ▶ Not maintaining the technical file — re-audit issues

Key Takeaways

Key takeaways from this lecture

Plan compliance early

Budget time + money up-front.

Pre-certified modules help

Wireless cert savings.

Pre-compliance saves money

Catch failures cheap.

Region-specific certs

CE / FCC / BIS / UL each.

Substance restrictions

RoHS, REACH, WEEE.

Maintain technical file

Required for 10 years.

Recommended Resources

Compliance learning paths

- ▶ EMC FastPass YouTube — practical compliance walkthroughs
- ▶ TUV / UL / Intertek free webinars
- ▶ CE marking guide on European Commission website
- ▶ FCC OET (Office of Engineering and Technology) free guides
- ▶ BIS India website + Compulsory Registration Scheme docs
- ▶ WPC ETA portal — wireless approval process
- ▶ Local labs: SAMEER, ICAT, ERTL, Lakshmi Labs
- ▶ Henry Ott — EMC Engineering (foundational)

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ EMC FastPass YouTube
- ▶ European Commission CE marking website (free)
- ▶ FCC OET guides (free)
- ▶ BIS India CRS guide
- ▶ WPC ETA portal
- ▶ Indian labs: SAMEER, ICAT, ERTL
- ▶ Local consultants for hand-holding
- ▶ Trade associations: ELCINA, MAIT (India)

Thank you • Build something this week.